Vector Calculus and Complex Analysis Homework 11

1. Evaluate the following integrals with counterclockwise contour:

(a) $\oint_{|z|=2} \frac{z^n}{z-1} dz, \quad n \geq 0$

(b) $\oint_{|z|=1} \frac{z^n}{z-2} dz, \quad n \geq 0$

(c) $\oint_{|z|=1} \frac{\sin z}{z} dz$

(d) $\oint_{|z-1-i|=5/4} \frac{\mathrm{Log}\, z}{(z-1)^2} dz$

(e) $\oint_{|z|=1} \frac{dz}{z^2(z^2-4)e^z}$

(f) $\oint_{|z-1|=2} \frac{dz}{z(z^2-4)e^z}$

Solution

(a) The nominator has one zero at $z = 1$ inside $|z| = 2$, thus by by Cauchy´s theorem,

$$\oint_{|z|=2} \frac{z^n}{z-1} dz = \frac{2\pi i}{0!} [z^n]_{z=1} = 2\pi i.$$

(b) The nominator has no zeros inside $|z| = 1$, thus by by Cauchy´s theorem,

$$\oint_{|z|=1} \frac{z^n}{z-2} dz = 0.$$

(c) The nominator has one zero at $z = 0$ inside $|z| = 1$, thus by by Cauchy´s theorem,

$$\oint_{|z|=1} \frac{\sin z}{z} dz = \frac{2\pi i}{0!} [\sin z]_{z=0} = 0.$$

(d) Let f be the principal branch of the logarithm defined on the slit plane, that is, $f(z) = \operatorname{Log} z$ for all $z \neq 0$ with $\operatorname{Arg} z \neq -\pi$. Then f is analytic on a domain $|z-1-i| \leq 5/4$. The nominator has one double zero at $z = 1$ inside $|z-1-i| < 5/4$, thus by by Cauchy´s theorem.

$$\oint_{|z-1-i|=5/4} \frac{\operatorname{Log} z}{(z-1)^2} dz = \frac{2\pi i}{1!} \left[\frac{d}{dz} (\operatorname{Log} z) \right]_{z=1} = 2\pi i.$$

(e) The nominator has one double zero at $z = 0$ inside $|z| = 1$, thus by by Cauchy´s theorem,

$$\oint_{|z|=1} \frac{dz}{z^2 (z^2-4) e^z} = \frac{2\pi i}{1!} \left[\frac{d}{dz} \left(\frac{e^{-z}}{z^2-4} \right) \right]_{z=0} = \frac{\pi i}{2}.$$

(f) The nominator has one zero at $z = 0$, and one zero at $z = 2$ inside $|z-1| = 2$, thus by by Cauchy´s theorem,

$$\oint_{|z-1|=2} \frac{dz}{z (z^2-4) e^z} = \frac{2\pi i}{0!} \left[\frac{1}{(z^2-4) e^z} \right]_{z=0} + \frac{2\pi i}{0!} \left[\frac{1}{z (z+2) e^z} \right]_{z=2} = -\frac{\pi i}{2} + \frac{\pi i}{4e^2}.$$

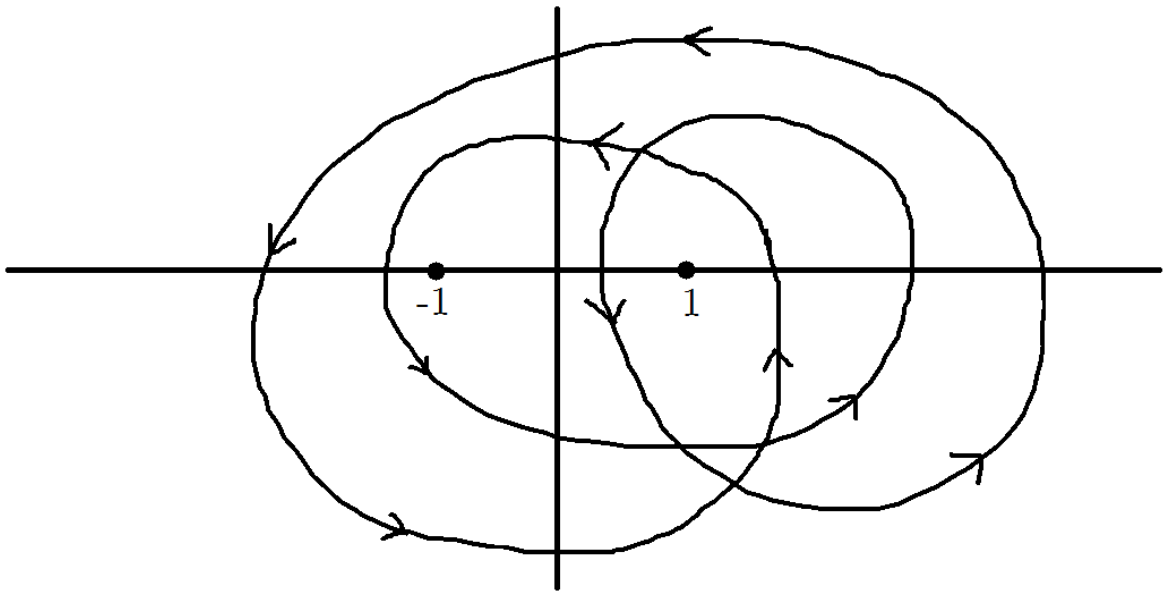
2. Let f and g be analytic inside and on the simple loop Γ. Prove that if $f(z) = g(z)$ for all z on Γ, then $f(z) = g(z)$ for all z inside Γ.

Solution

For z_0 inside Γ,

$$f(z_0) = \frac{1}{2\pi i} \oint_\Gamma \frac{f(z)}{z - z_0} dz = \frac{1}{2\pi i} \oint_\Gamma \frac{g(z)}{z - z_0} dz = g(z_0)$$

3. Let C be the curve in below figure. Calculate $\int_C \frac{e^z}{z^3 - z^2 - z + 1} dz$.



Solution

$$\begin{aligned}
\int_C \frac{e^z}{z^3-z^2-z+1}\,dz &= 3\int_{C_1} \frac{e^z}{(z-1)^2(z+1)}\,dz + 2\int_{C_{-1}} \frac{e^z}{(z-1)^2(z+1)}\,dz \\
&= 6\pi i \frac{d}{dz}\left(\frac{e^z}{z+1}\right)\Bigg|_{z=1} + 4\pi i\left(\frac{e^z}{(z-1)^2}\right)\Bigg|_{z=-1} \\
&= 6\pi i\left(\frac{(z+1)e^z - e^z}{(z+1)^2}\right)\Bigg|_{z=1} + 4\pi i\left(\frac{e^z}{(z-1)^2}\right)\Bigg|_{z=-1} \\
&= 6\pi i\left(\frac{e}{4}\right) + 4\pi i\left(\frac{e^{-1}}{4}\right) \\
&= \frac{3e+2e^{-1}}{2}\pi i
\end{aligned}$$

4. Suppose that *f* is entire and that $|f(z)| \le |z|^2$ for all sufficiently large values of |*z*|, say $|z| > r_0$. Prove that *f* must be a polynomial of degree at most 2. Generalize this result.

[HINT: Show by the Cauchy estimates that for *R* sufficiently large $|f^{(3)}(z_0)| \le \dfrac{3!(R+|z_0|)^2}{R^3}$, and thereby conclude that $f^{(3)}$ vanishes everywhere.]

Solution

Fix z_0 and choose $R > |z_0| + r_0$. Then the circle $|z - z_0| = R$ lies outside the circle $|z| = r_0$, and one can apply Cauchy's estimate on the circle $|z - z_0| = R$ to get

$$|f^{(n)}(z_0)| \leq \frac{n! \max_{|z-z_0|=R} |z|^2}{R^n} = \frac{n!(|z_0| + R)^2}{R^n}$$

For $n \geq 3$, letting R approach ∞ yields $f^{(n)}(z_0) = 0$, so that $f^{(n)} \equiv 0$. Hence f is a polynomial of degree at most 2.

In general, if f is entire and $|f(z)| \leq M|z|^n$ for $|z| > r_0$, where n is a nonnegative integer, then f must be a polynomial of degree at most n.

5. Find the maximum and minimum absolute value of $z^2 - z$ in the disc: $|z| \leq 1$.

Solution

The maximum of $\left|z^2 - z\right|$ must occur on the boundary of the disk ($|z| = 1$).

The latter can be parametrized $z = e^{it}$, $0 \leq t \leq 2\pi$; whence

$$\left|z^2 - z\right|^2 = \left|z - 1\right|^2 = (e^{it} - 1)(e^{-it} - 1) = 1 - e^{it} - e^{-it} + 1 = 2 - 2\cos t.$$

The maximum and minimum value of $\left|z^2 - z\right|$ is 4 and 0 respectively.